Chapter 3
Analog Transmission

Chapter Outline

3.1 Introduction

"Modulate" means to regulate or adjust. In communication, it means to regulate some parameter of a high-frequency carrier wave with a lower frequency information signal

3.2 Signal Conversion

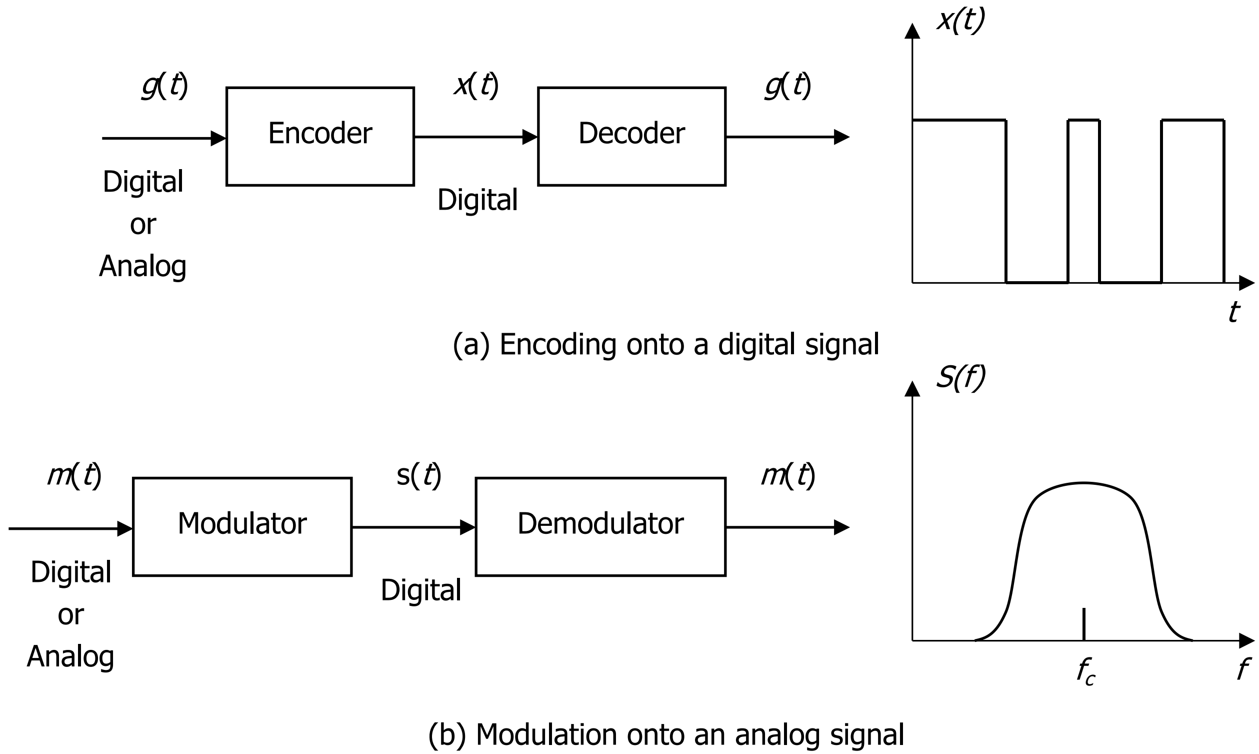


Figure 3.1: Encoding and modulation techniques

Types of Data to Signal Conversion

1. Digital data to digital signal
2. Analog data to digital signal
3. Digital data to analog signal
4. Analog data to analog signal

3.3 Analog Data Analog Signal

Principal reasons for analog modulation of analog signals:

1. **A higher frequency may be needed for effective transmission. For unguided transmission, it is virtually impossible to transmit baseband signals; the required antennas would be many kilometers in diameter.**
2. **Modulation permits frequency division multiplexing.**

Mathematical Expression of wave:

The mathematical expression for a sinusoidal carrier wave is -

$$e = E_c \sin(\omega_c t + \varphi)$$
$$e = E_c \sin(2\pi f_c t + \varphi)$$

Obviously the waveform can be varied by any of its following three factors or parameter-

E_c - the amplitude
f_c - the frequency
φ - the phase

3.3.1 Amplitude Modulation (AM)

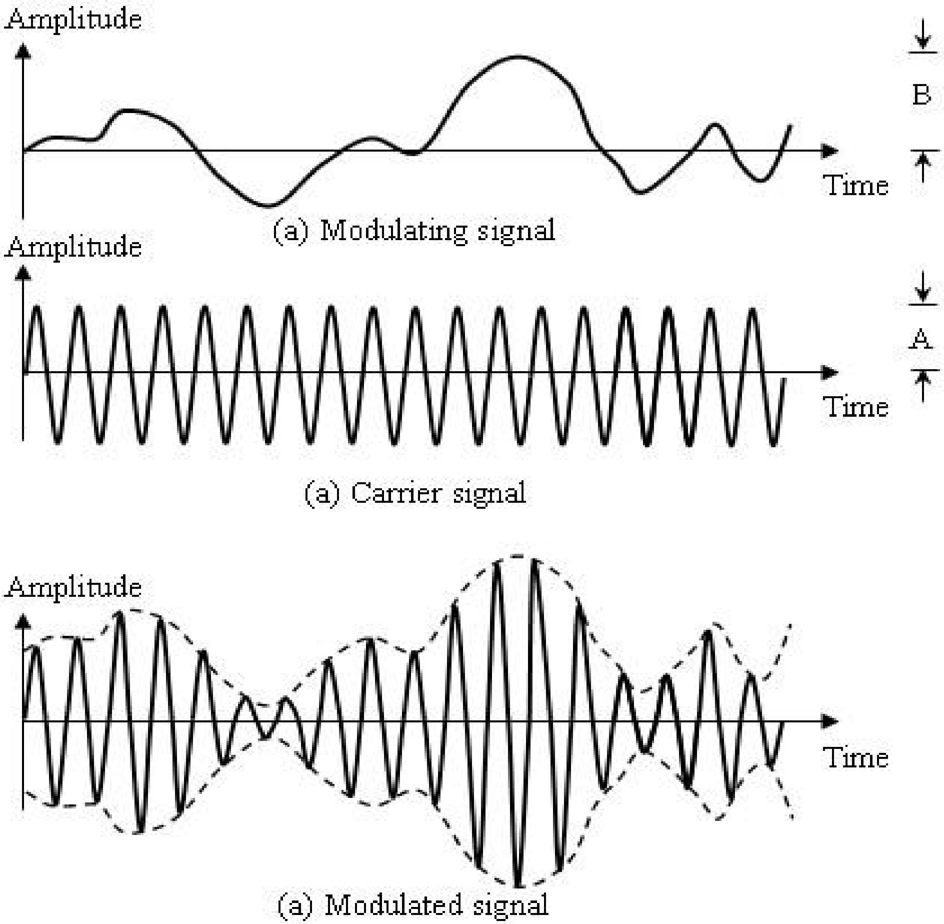


Figure 3.2: Amplitude modulation

Percent Modulation

Percent modulation, m, indicates the degree to which the AF signal modulates the carrier wave

$$m = \frac{\max imum\ value\ of\ signal\ wave}{\max imum\ value\ of\ carrier\ wave} \times 100$$

$$= \frac{signal\ amplitude}{carrier\ amplitude} \times 100$$

$$= \frac{B}{A} \times 100$$

$$m = \frac{E_{c(\max)} - E_{c(\min)}}{E_{c(\max)} + E_{c(\min)}} X100$$

Effect of modulation Index

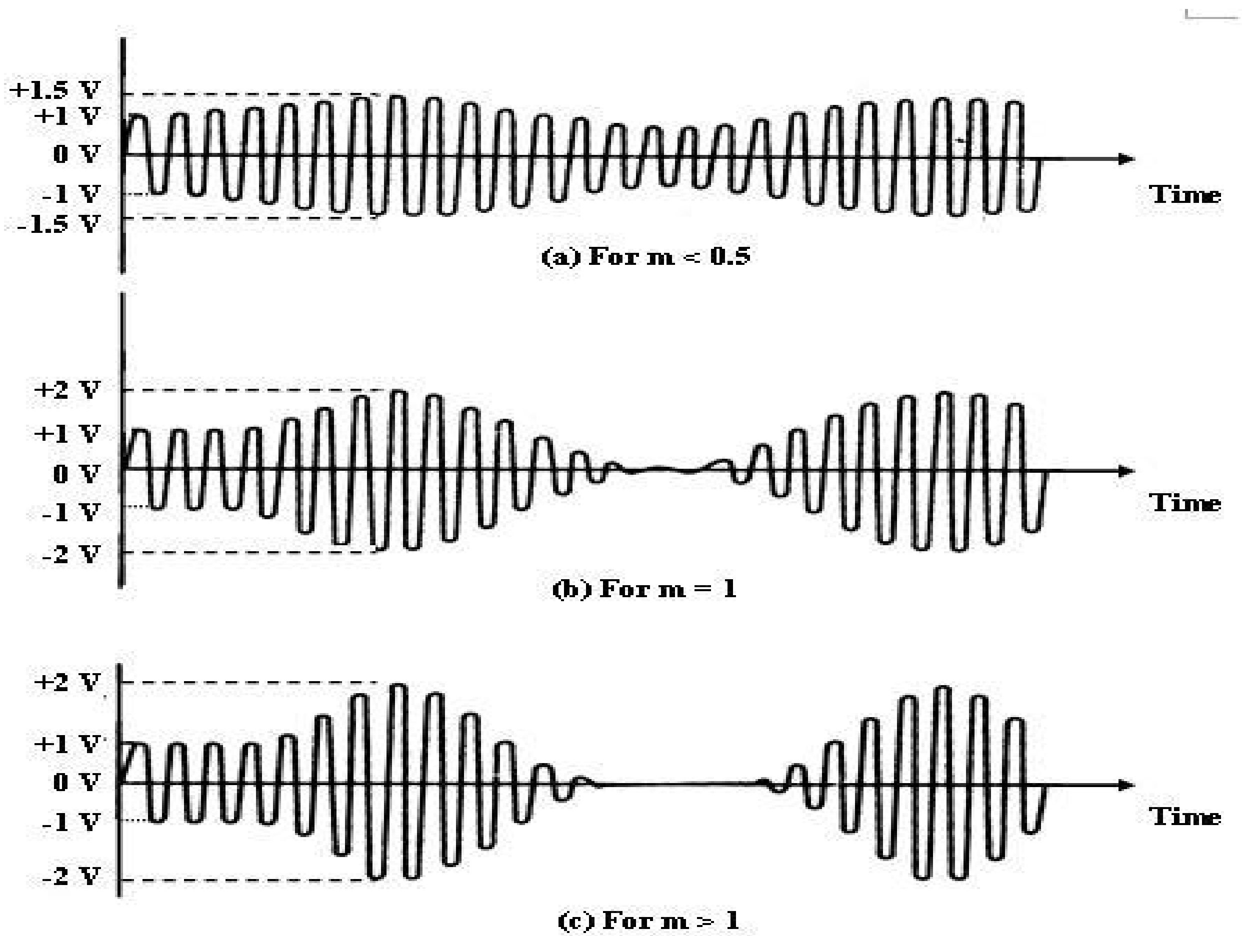


Figure 3.3: Amplitude modulation of various indexes

Amplifier Analog Modulation

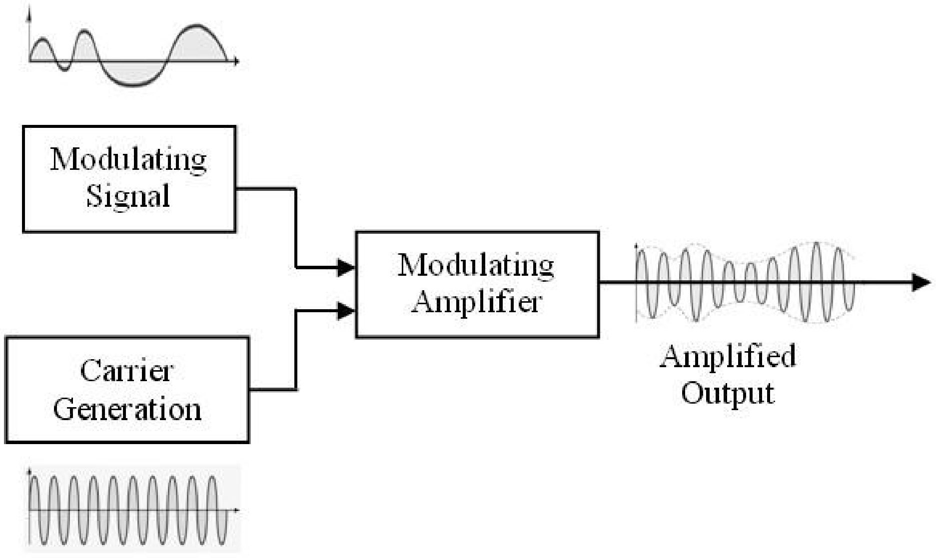


Figure 3.4: Methods of amplitude modulation

Amplifier Analog Modulation

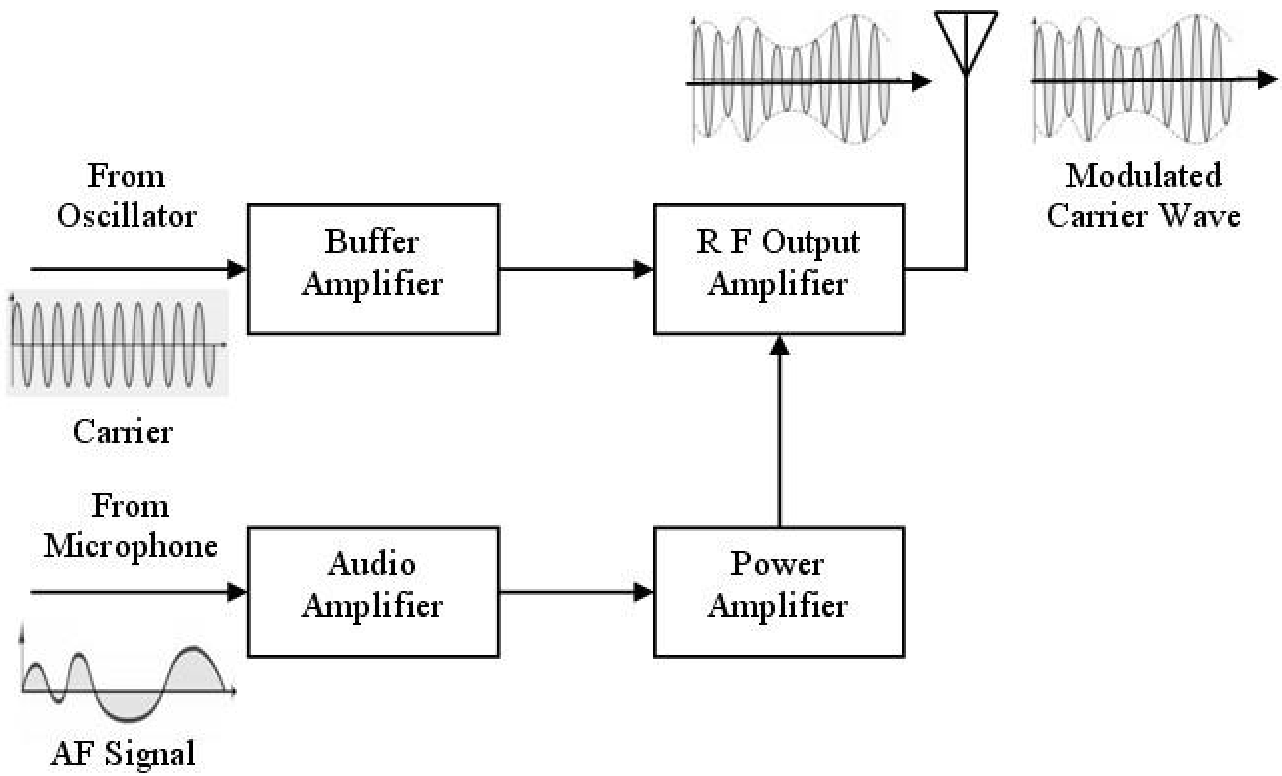


Figure 3.5: Block diagram of a typical AM transmitter

Amplifier Detection

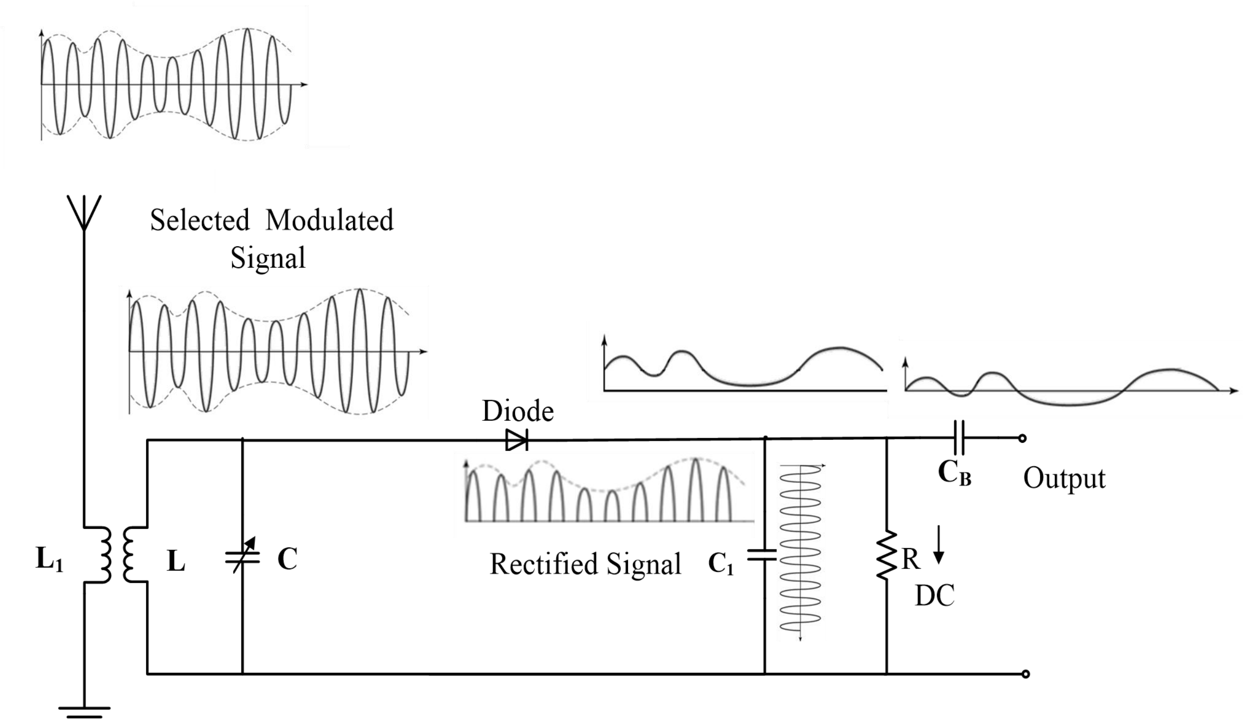


Figure 3.6: Amplitude detection

AM Bandwidth

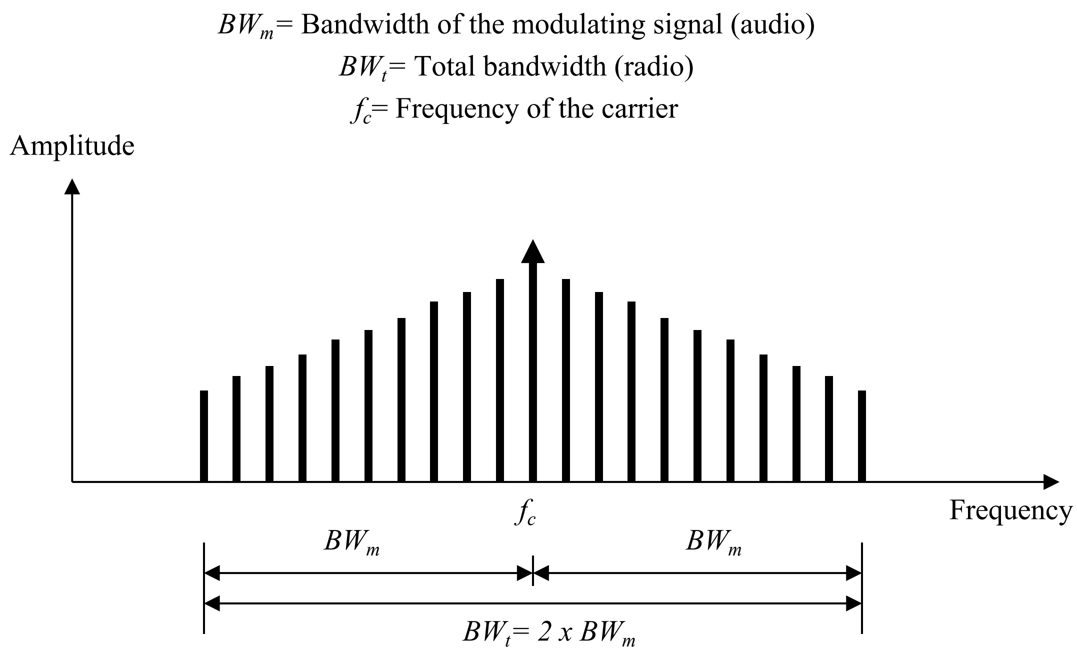


Figure 3.7: Amplitude modulation bandwidth

Amplitude modulation band allocation

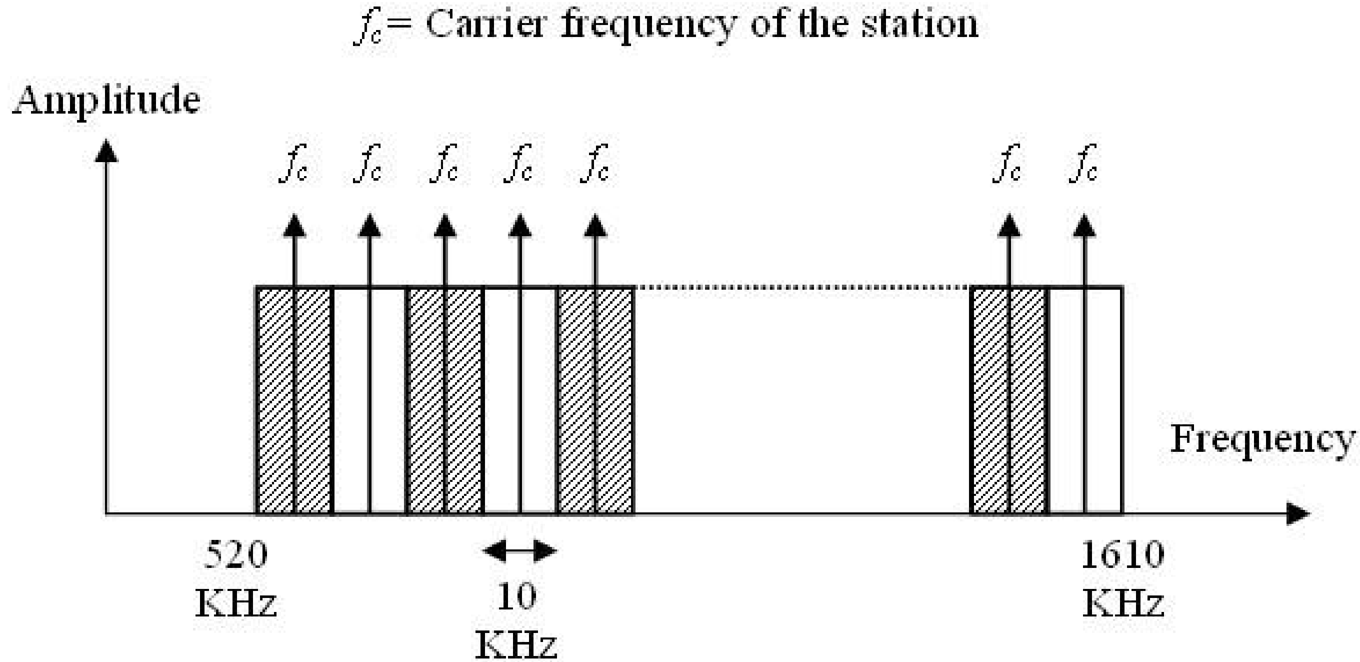


Figure 3.8: Amplitude modulation band allocation

3.3.2 Frequency Modulation (FM)

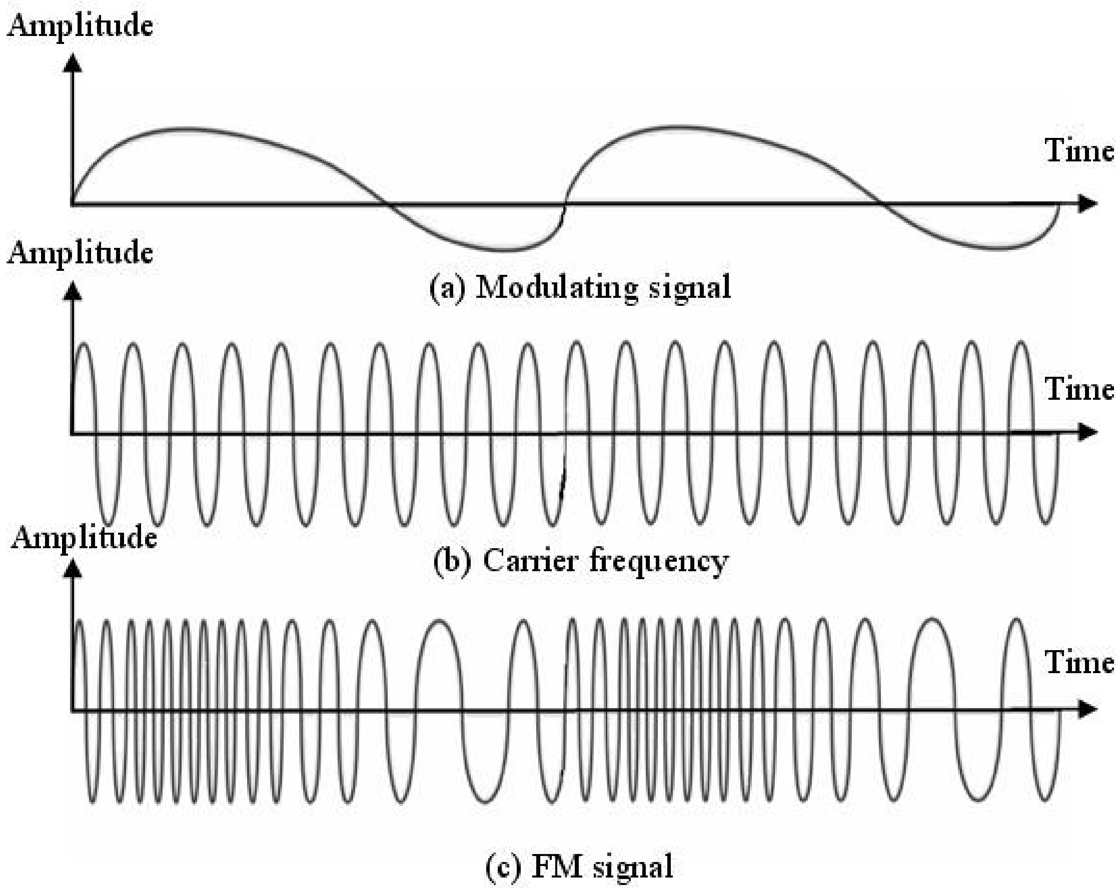


Figure 3.9: Frequency modulation

Mathematical Analysis of FM Signal

The instantaneous frequency *f* for a frequency modulated signal is given by

$$f = f_c\left(1 + KE_m \cos \omega_m t\right)$$

where, K = proportionality constant

f_c = unmodulated carrier frequency

f_m = carrier frequency of the modulating signal

E_m = maximum value of the modulating voltage

ω_c = unmodulated angular frequency of the carrier

ω_m = angular frequency of the modulating signal

Considering m_f (modulation index) the a frequency modulated signal can be defined as -

$$e = E_c \sin\left(\omega_c t + m_f \sin \omega_m t\right)$$

FM Bandwidth

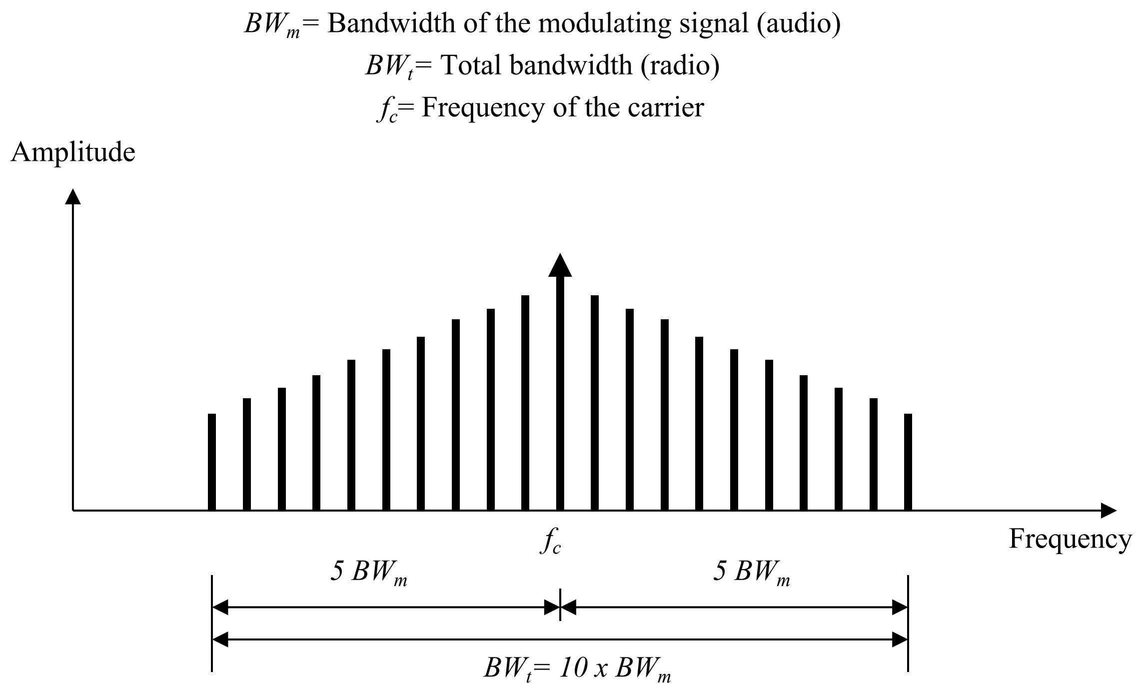


Figure 3.10: Frequency modulation bandwidth

FM Bandwidth

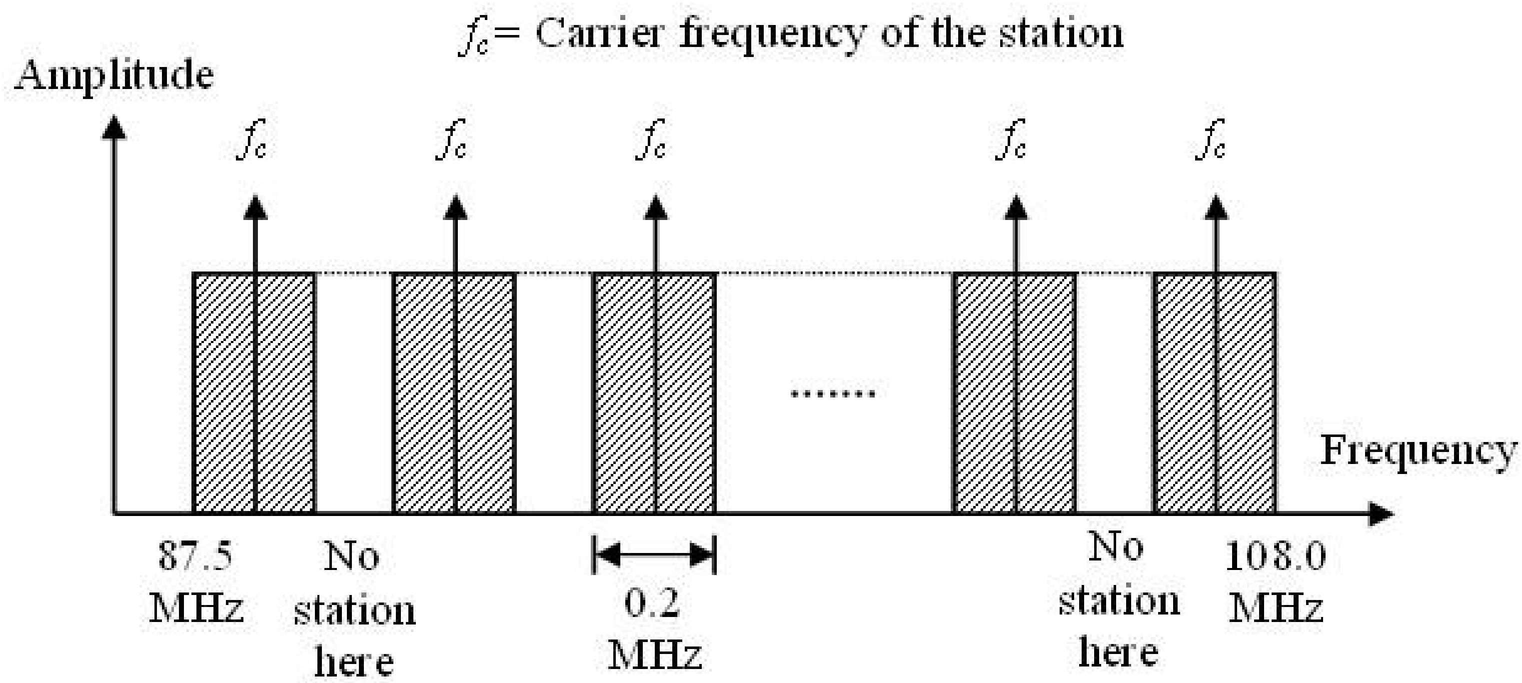


Figure 3.11: Frequency modulation band allocation

3.3.3 Phase Modulation (PM)

Phase modulation and frequency modulation are very closely related, and infact frequency modulation can be very easily obtained from phase modulation by the so called Armstrong method. Phase modulation and frequency modulation are basically two types of angle modula-tion.

The expression for a phase modulated wave will be-

$$e = E_c \sin\left(\omega_c t + \varphi_m \sin \omega_m t\right)$$

3.4 Digital Data, Analog Signal

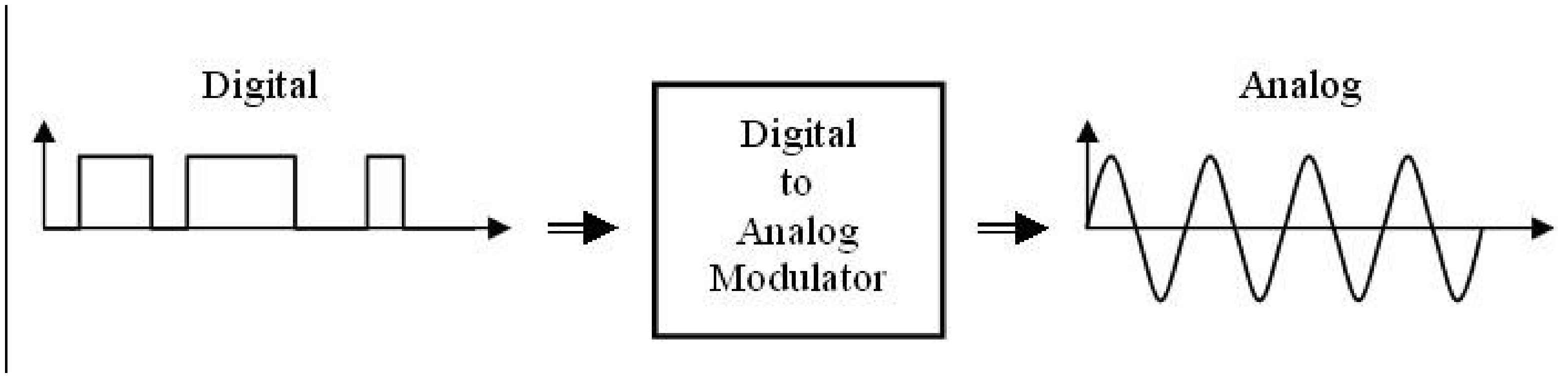


Figure 3.12: Digital to analog modulation

Note

Digital to analog modulation is the technique to convert digital data to an analog signal.

Types of digital data to analog modulation

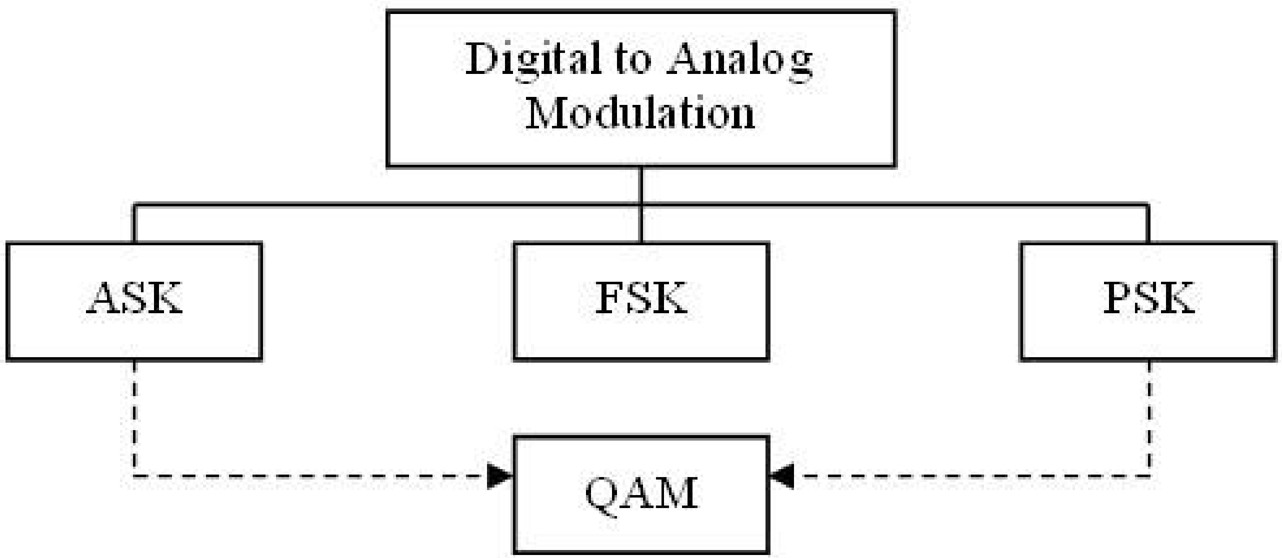


Figure 3.13: Types of digital data to analog modulation

3.4.1 Amplitude Shift Keying (ASK)

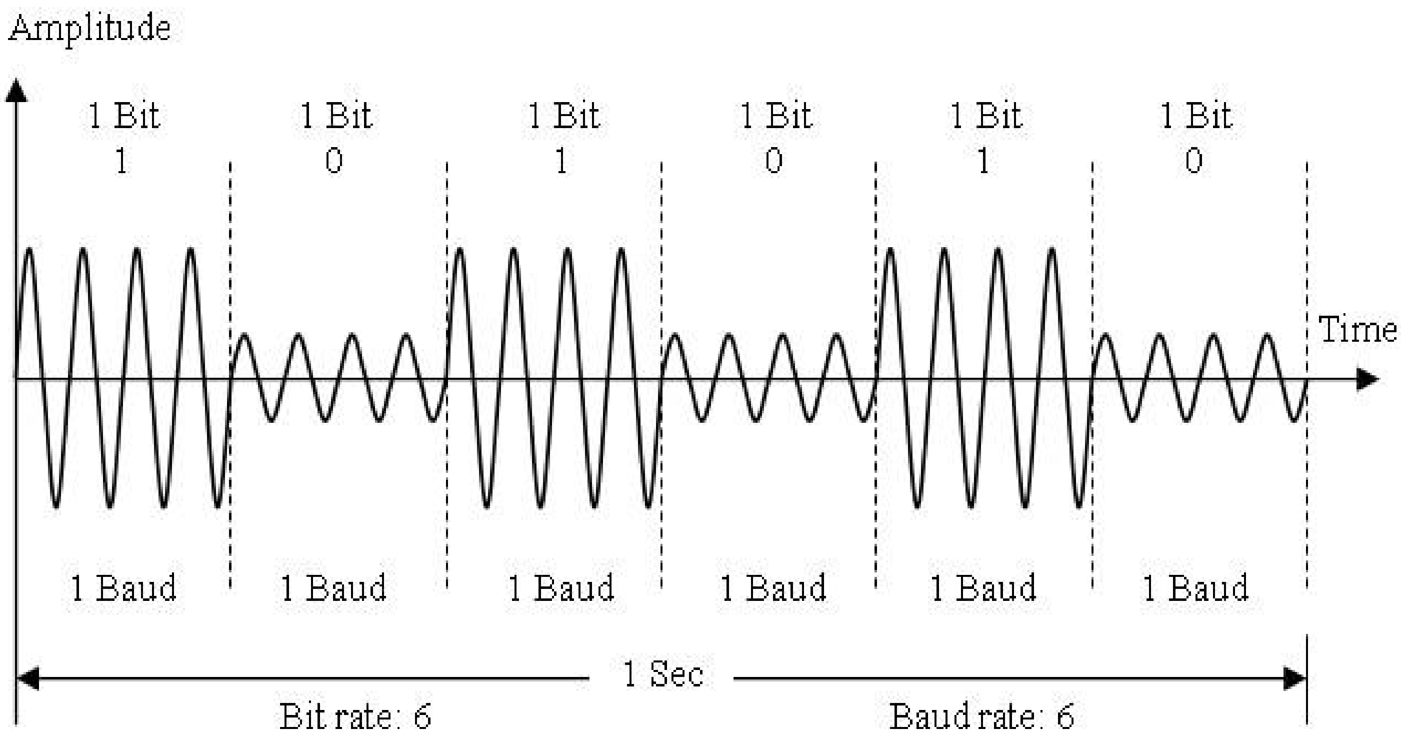


Figure 3.14: Amplitude shift keying

Note

Noise usually affects the amplitude; there-fore, thus ASK is most affected by noise.

Amplitude Shift Keying (Cont.)

On/Off Keying (OOK) is a popular ASK technique. In OOK, logic 0 is represented by the absence of a carrier. This can save the required energy to transmit information.

In mathematical terms the ASK modulated signal can be expressed as:

$$s(t) = \begin{cases} A_c \cos(2\pi f_c t), & \text{for bit } 1 \\ 0, & \text{for bit } 0 \end{cases}$$

Bandwidth for ASK

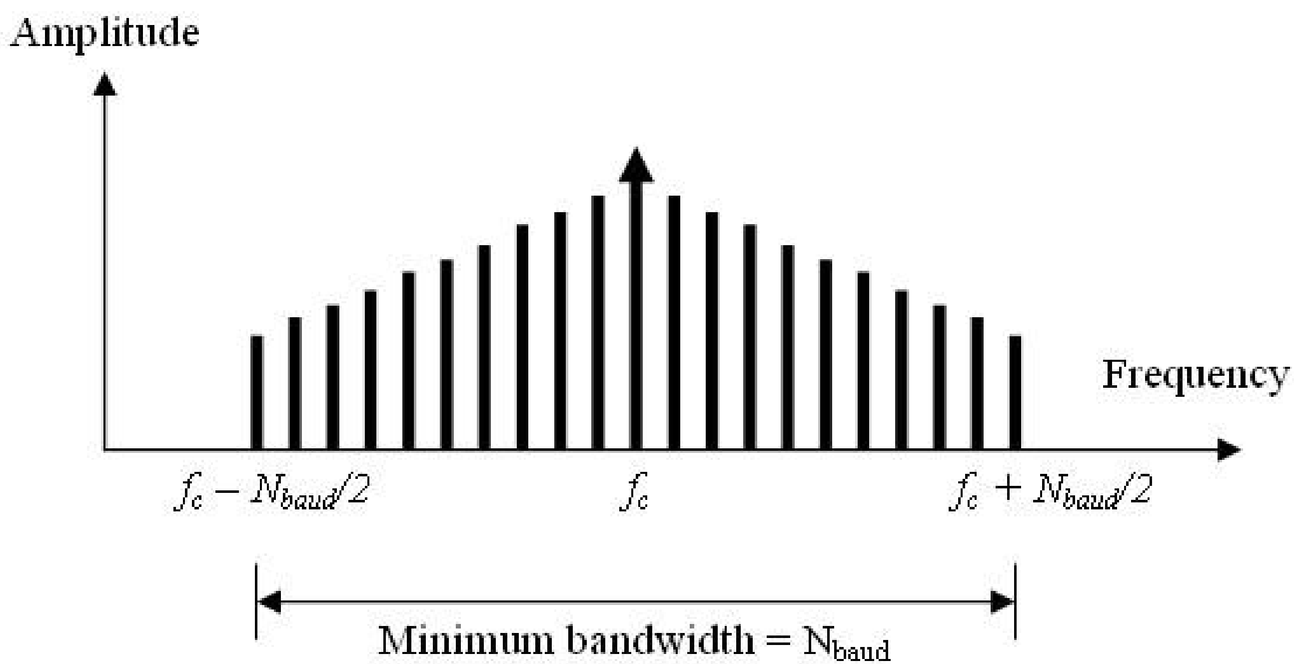


Figure 3.15: Relationship between Baud rate and bandwidth in ASK

3.4.2 Frequency Shift Keying (FSK)

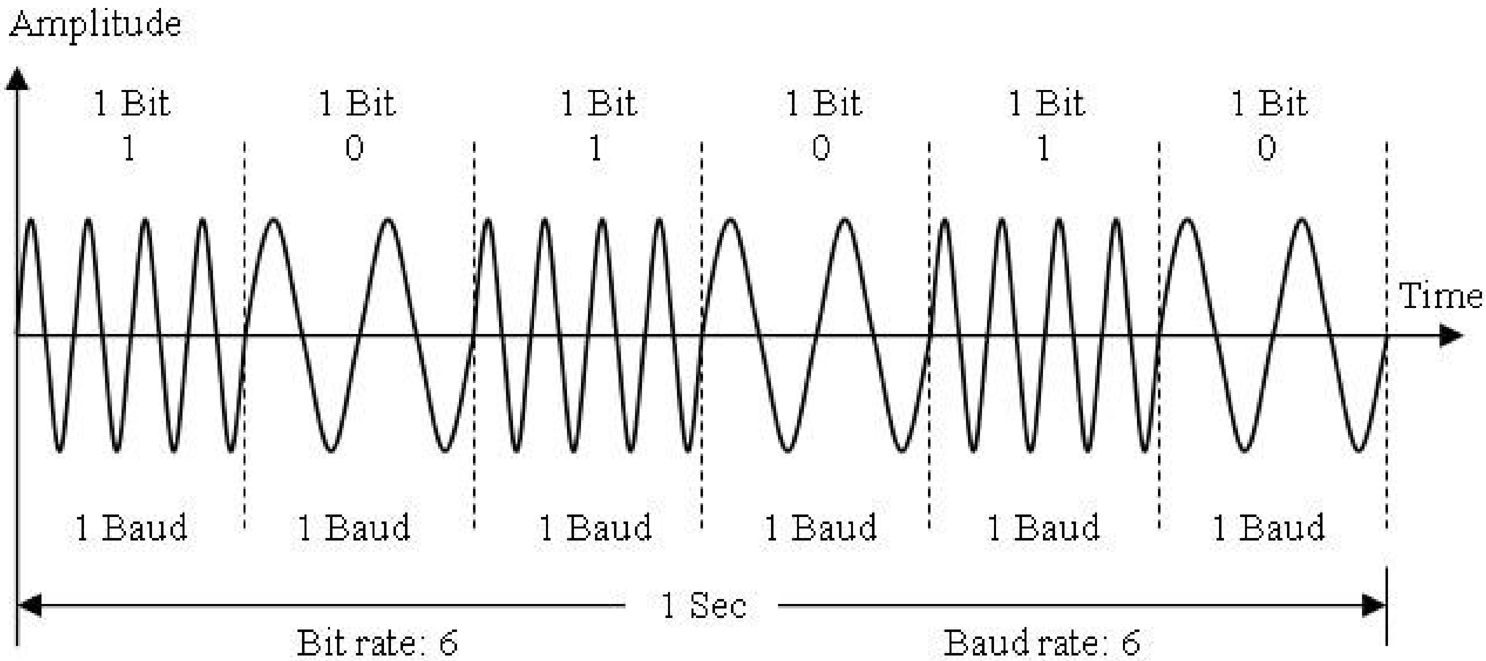


Figure 3.16: Frequency shift keying

Frequency Shift Keying (cont.)

In FSK system, two sinusoidal waves of the same amplitude but different frequencies f_1 and f_2 are used to represent binary bits 1 and 0 respectively.

In mathematical terms the FSK modulated signal can be expressed as:

$$s(t) = \begin{cases} A_c \cos(2\pi f_1 t), & \textit{for bit } 1 \\ A_c \cos(2\pi f_2 t), & \textit{for bit } 0 \end{cases}$$

Bandwidth for FSK

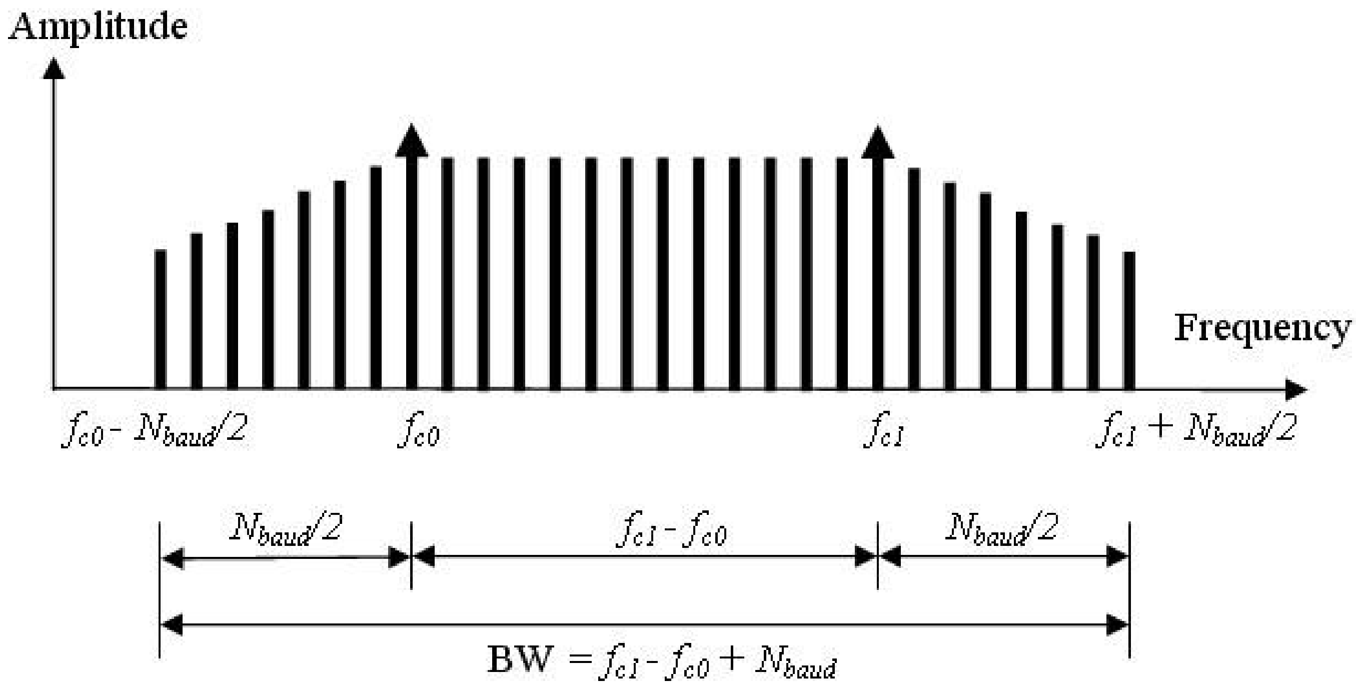


Figure 3.17: Relationship between Baud rate and bandwidth in FSK

3.4.3 Phase Shift Keying (PSK)

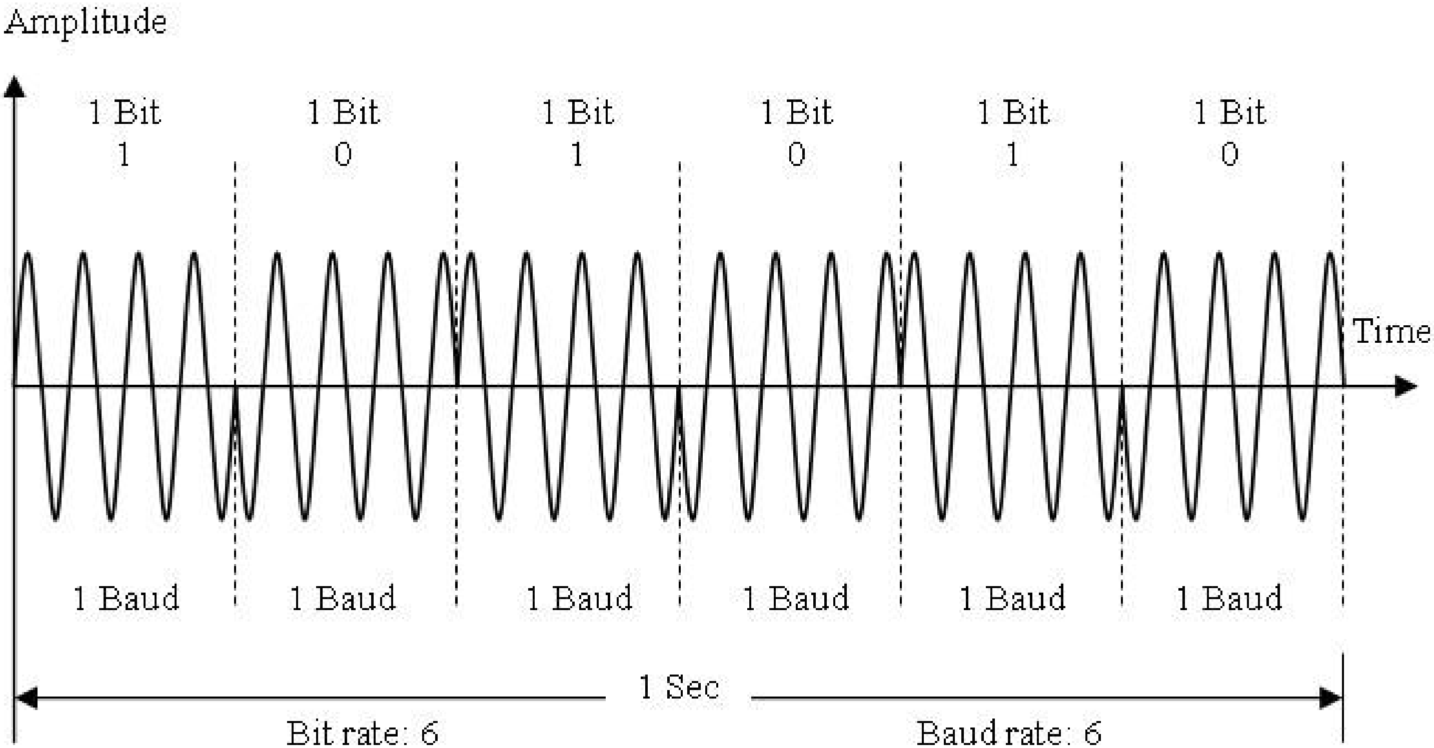


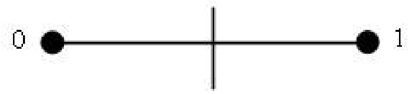
Figure 3.18: Phase shift keying

Phase Shift Keying (cont.)

In PSK system, two sinusoidal waves of the same amplitude and frequency *fc* but phases are 0 and π are used to represent binary bits 1 and 0 respectively.

In mathematical terms, the PSK modulated signal can be expressed as .

$$s(t) = \begin{cases} A_c \cos 2\pi f_c t, & 0 \le t \le T, \quad for\, 1 \\ -A_c \cos 2\pi f_c t, & 0 \le t \le T, \quad for\, 0 \end{cases}$$



Constellation Diagram

Bit	Phase
1	0°
0	180°

Bits

Figure 3.19: PSK constellation

Quadrature Phase-shift Keying (QPSK)

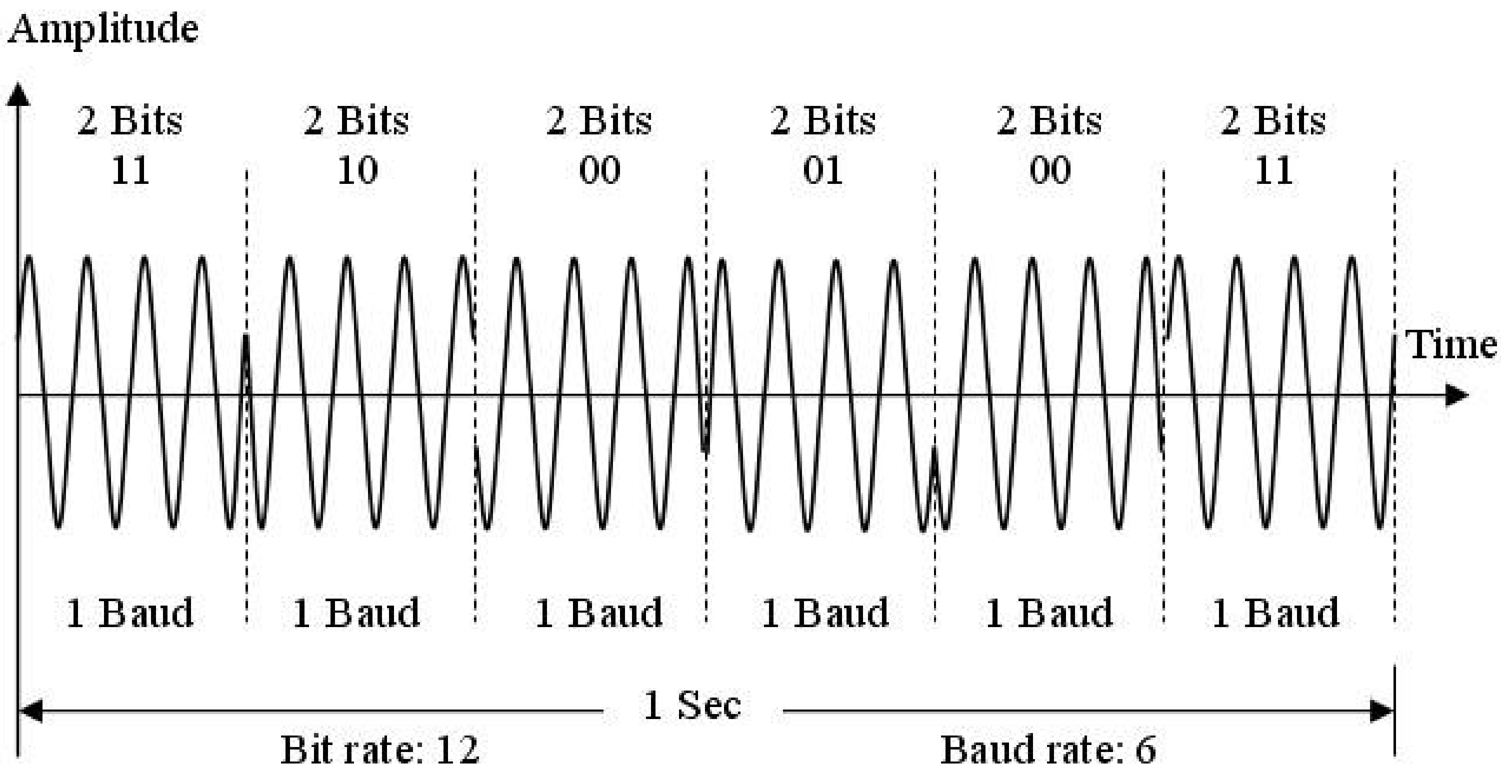


Figure 3.20: The 4-PSK method

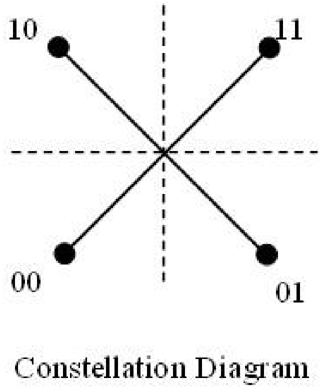
Quadrature Phase-shift Keying

In mathematical terms, the QPSK modulated signal can be expressed as-

$$s(t) = \begin{cases} A_c \cos\left(2\pi f_c t + \dfrac{\pi}{4}\right), & 0 \le t \le T, \quad \mathit{for}\ 11 \\ A_c \cos\left(2\pi f_c t + \dfrac{3\pi}{4}\right), & 0 \le t \le T, \quad \mathit{for}\ 10 \\ A_c \cos\left(2\pi f_c t + \dfrac{5\pi}{4}\right), & 0 \le t \le T, \quad \mathit{for}\ 00 \\ A_c \cos\left(2\pi f_c t + \dfrac{7\pi}{4}\right), & 0 \le t \le T, \quad \mathit{for}\ 01 \end{cases}$$

Quadrature Phase-shift Keying

The constellation diagram for the signal will be-



Dibit	Phase
11	45°
10	135°
00	225°
01	315°

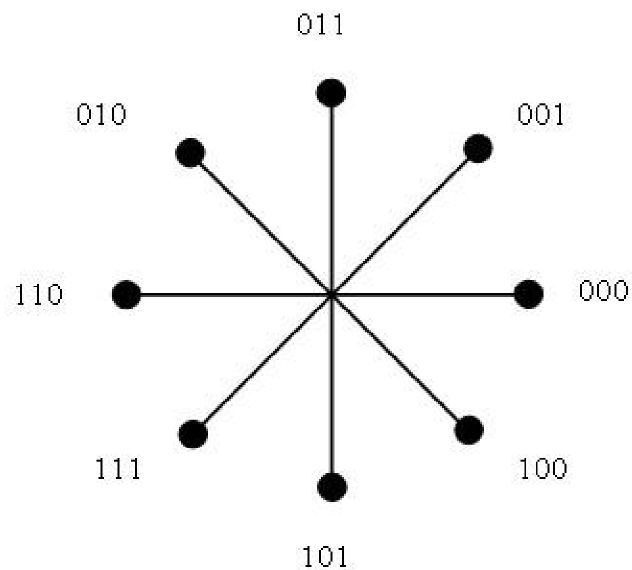
Dibits
(2 Bits)

Figure 3.21 The 4-PSK characteristics

Note

QPSK and 8-PSK are 2 and 3 times as efficient as 2-PSK respectively.

π/4 Phase shift Keying (8-PSK)



Constellation Diagram

Tribit	Phase
000	0°
001	45°
011	90°
010	135°
110	180°
111	225°
101	270°
100	315°

Tribits
(3 Bits)

Figure 3.22: The 8-PSK characteristics

Bandwidth for PSK

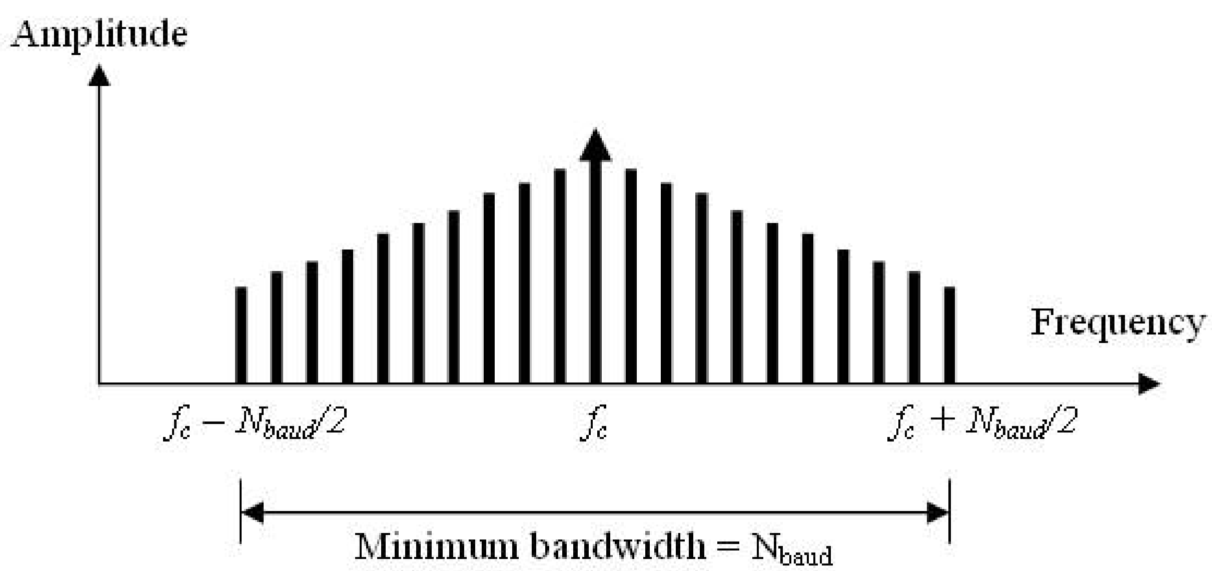


Figure 3.23: Relationship between Baud rate and bandwidth in PSK

3.4.4 Quadrature Amplitude Modulation (QAM)

Quadrature amplitude modulation is a combination of ASK and PSK such that a maximum contrast between each signal element (bit, dibit, tribit, and so on) is achieved.

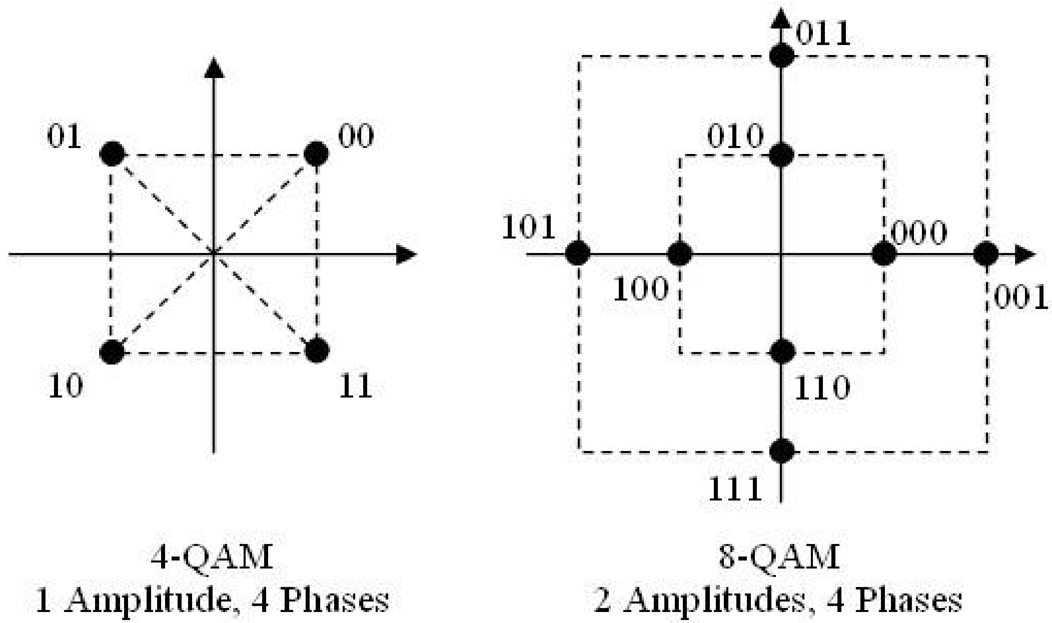


Figure 3.24: The 4-QAM and 8-QAM constellations

3.4.4 Quadrature Amplitude Modulation (QAM)

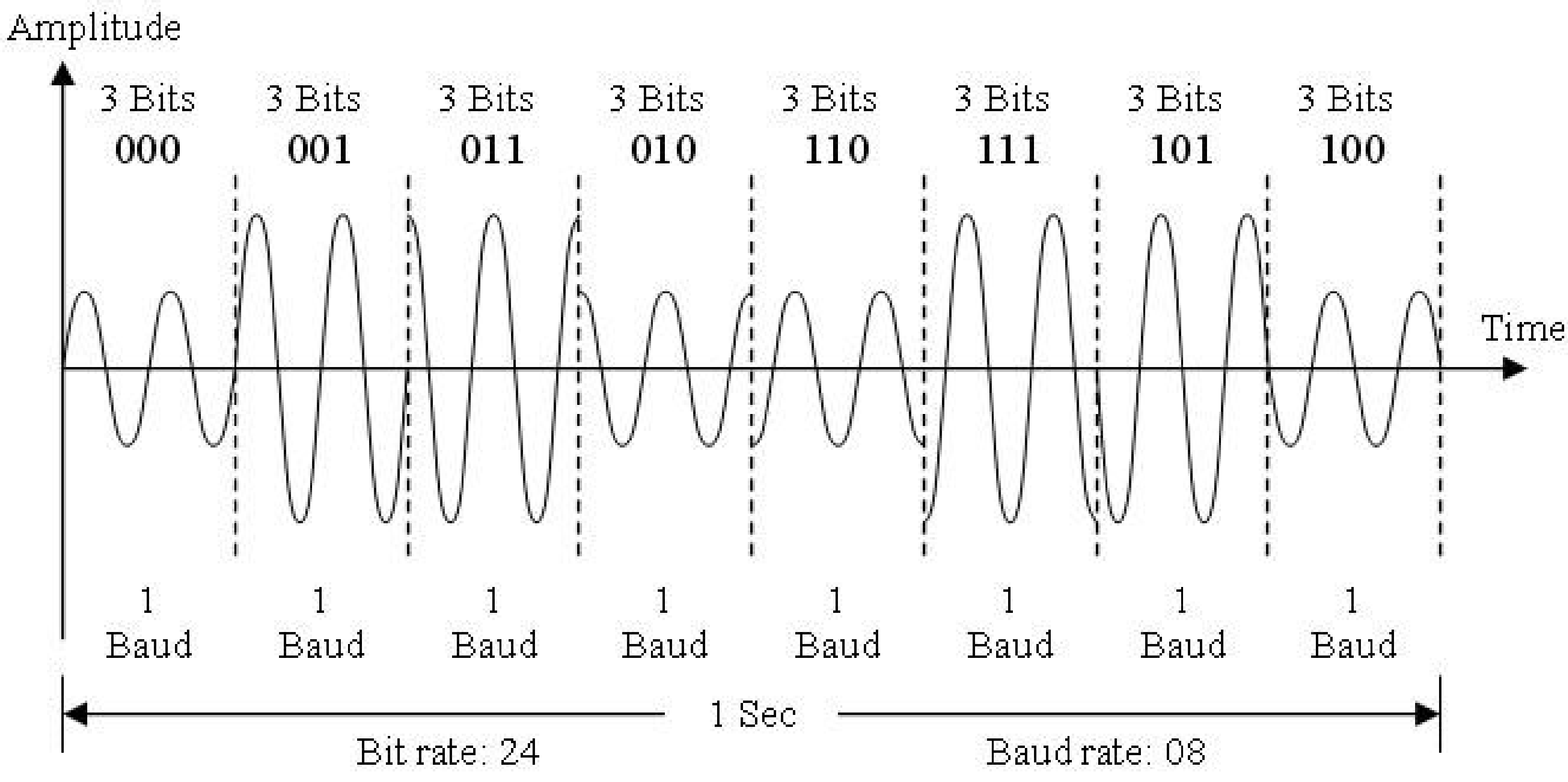


Figure 3.25: Time domain representation of 8-QAM signal

Various 16 QAM Constellation

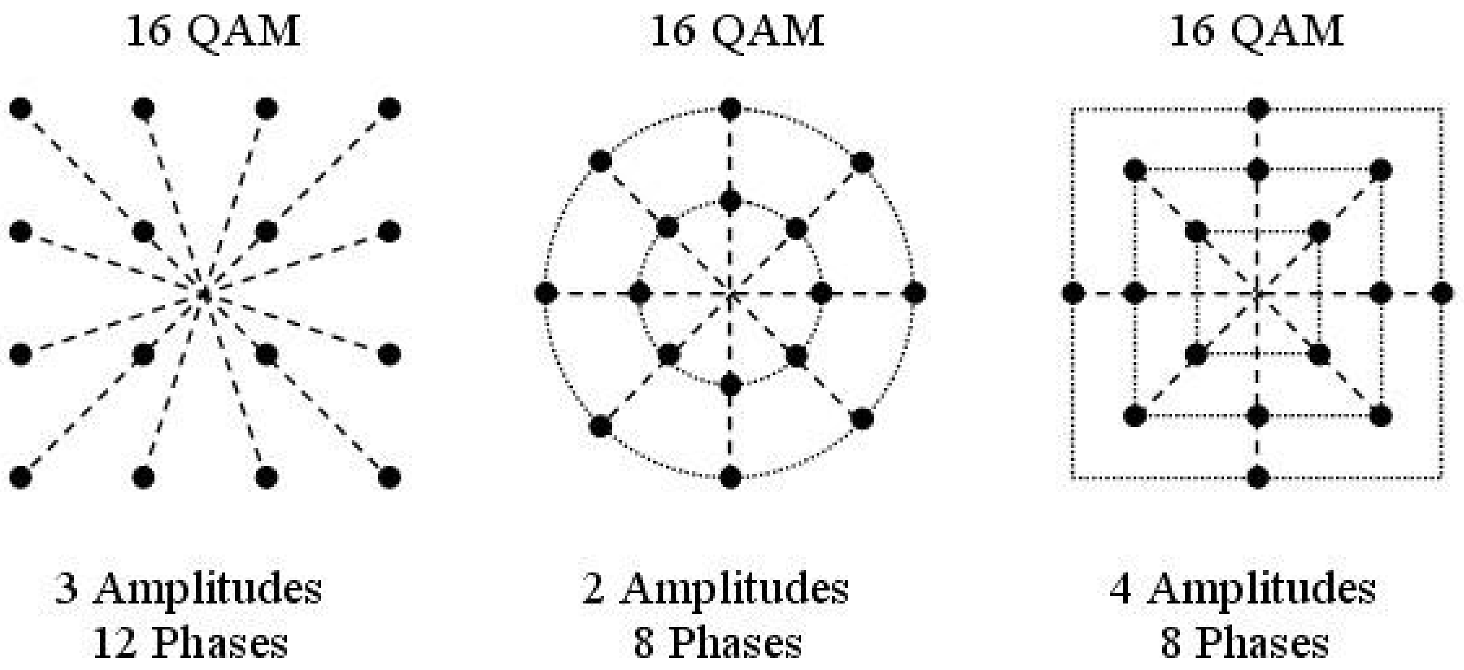


Figure 3.26: 16-QAM constellations

3.5 Sideband

In Amplitude Modulation (AM), a band of frequencies higher than or lower than the carrier frequency, contains energy as a result of the modulation process. The frequencies above the carrier frequency constitute the upper side band (USB); and those below the carrier frequency constitute the lower side band (LSB).

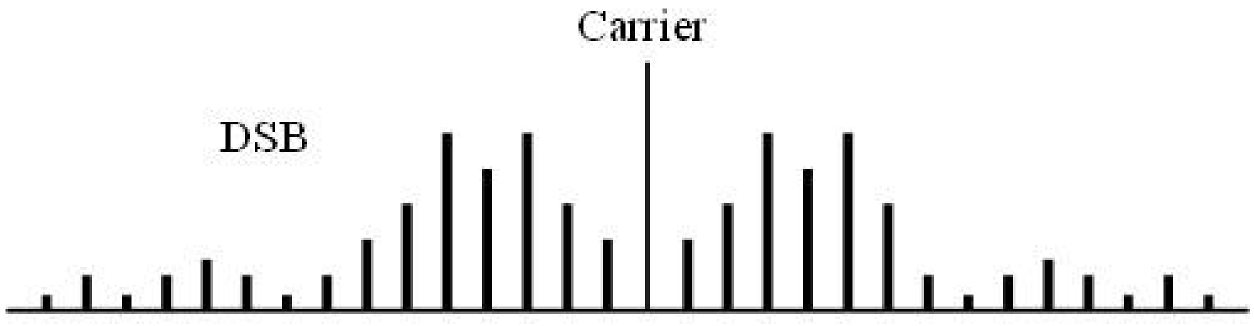


Figure 3.27: Frequency domain representation of DSB-AM

DSB-SC

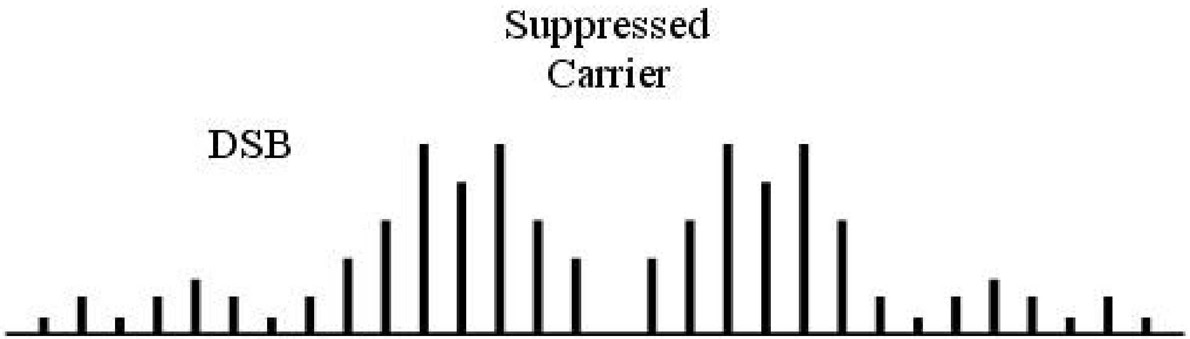


Figure 3.28: Frequency domain representation of DSB-SC

SSB-AM

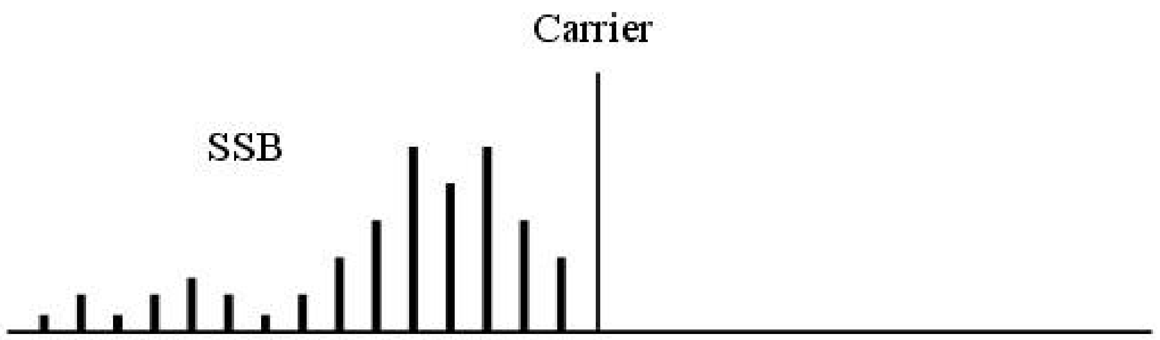


Figure 3.29: Frequency domain representation of SSB-AM

SSB-SC

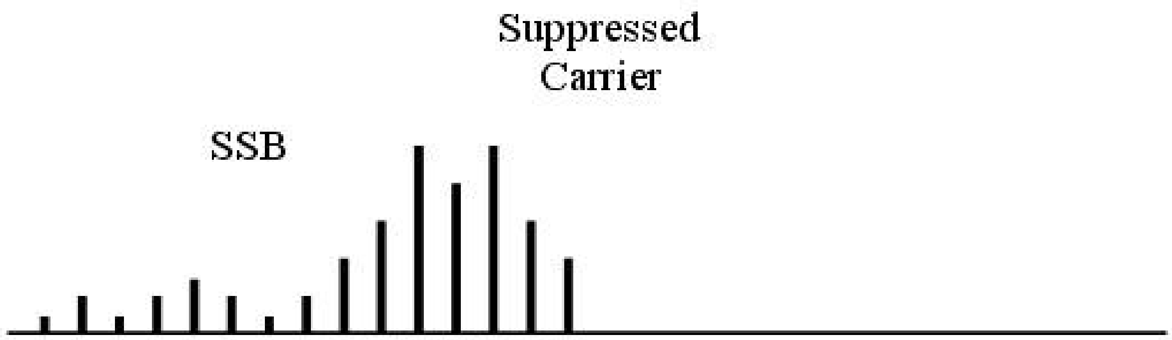


Figure 3.30: Frequency domain representation of SSB-SC

Vestigial Side Band (VSB)

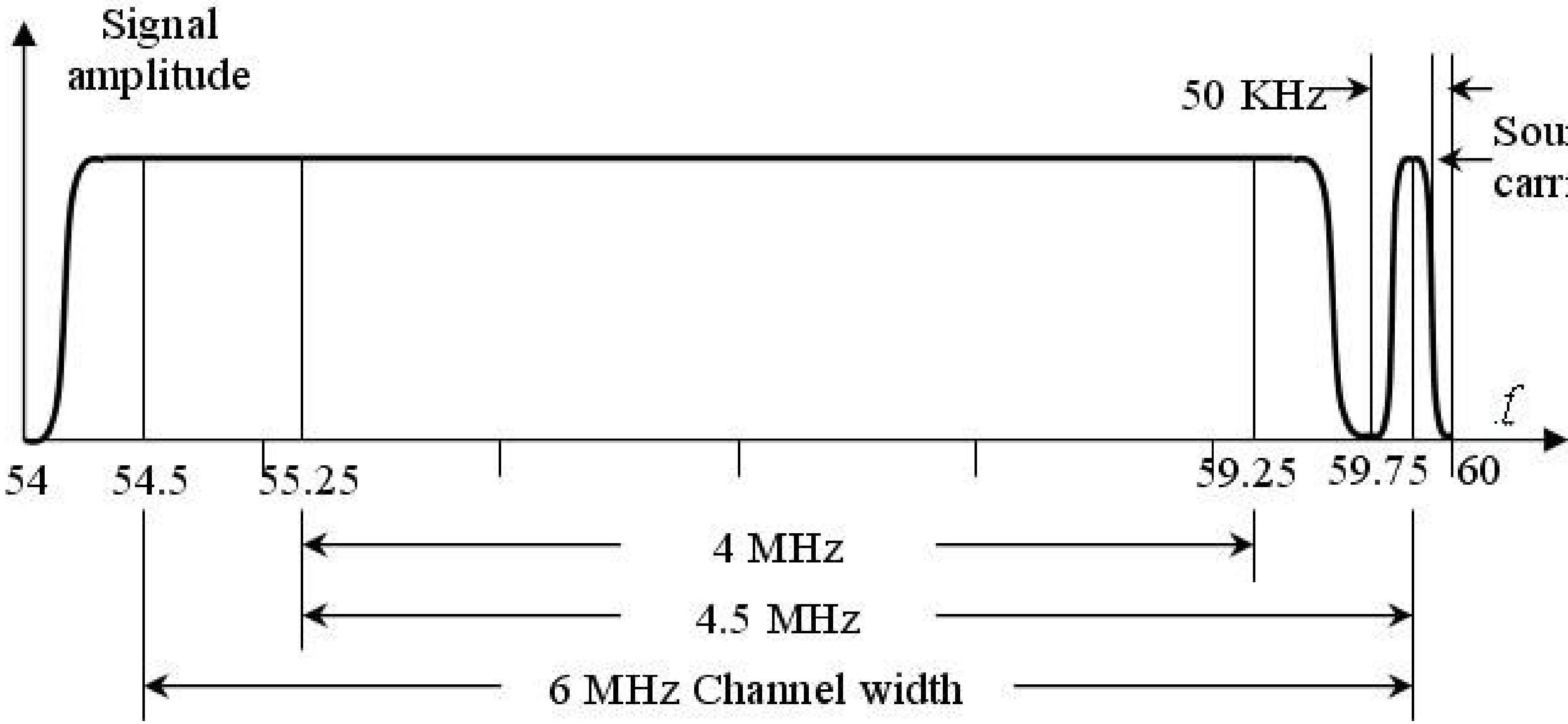


Figure 3.31: Allocated frequency range for picture carrier and sound carrier

3.6.1 Spread Spectrum Basic Principle

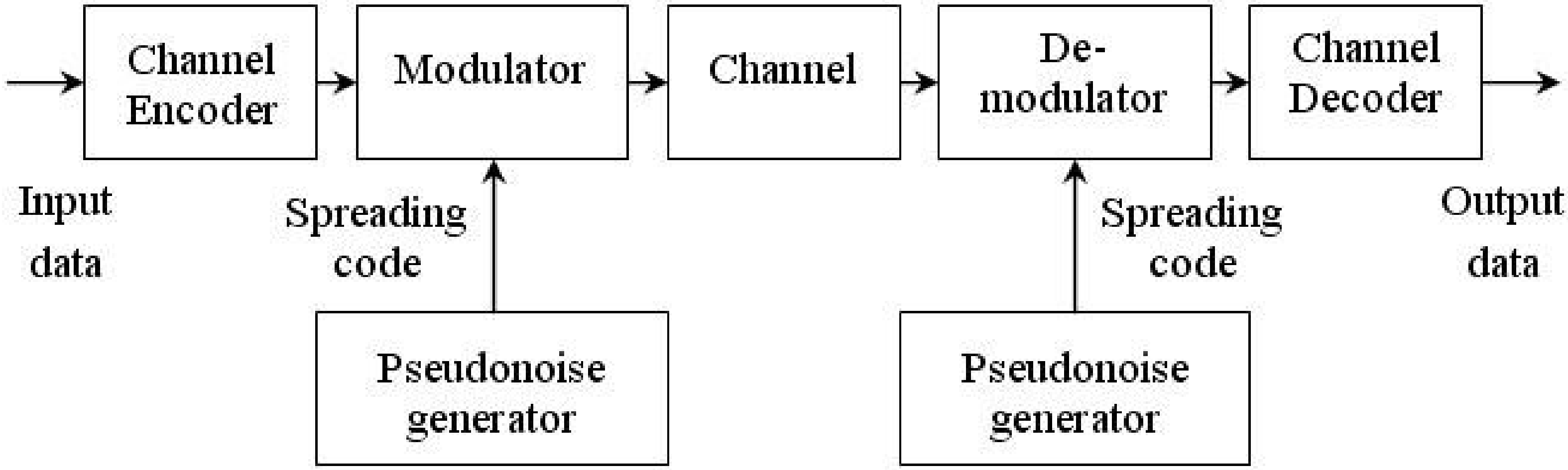


Figure 3.32: General model of spread spectrum digital communication system

3.6.2 Direct Sequence Spread Spectrum (DSSS)

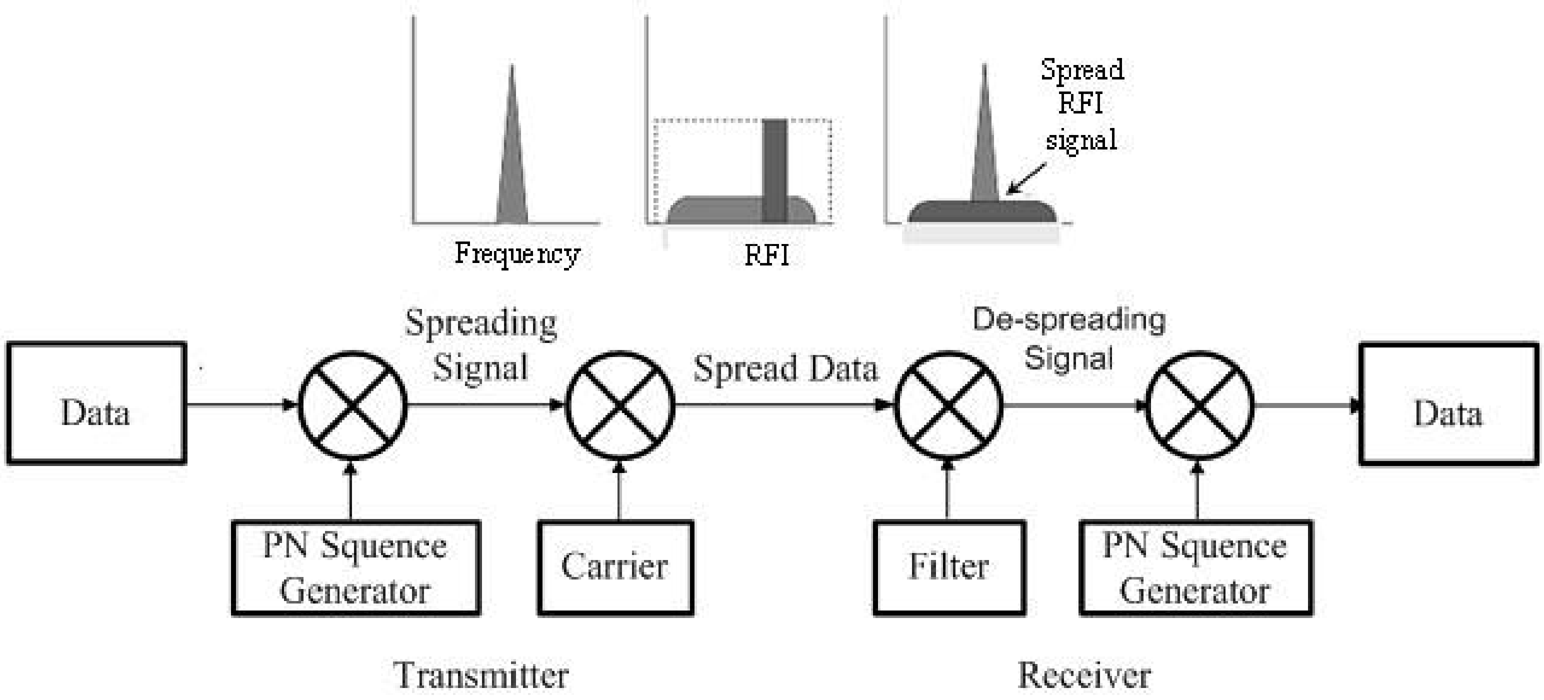


Figure 3.33: Direct Sequence Spread Spectrum

3.6.3 Frequency Hopping Spread Spectrum (FHSS)

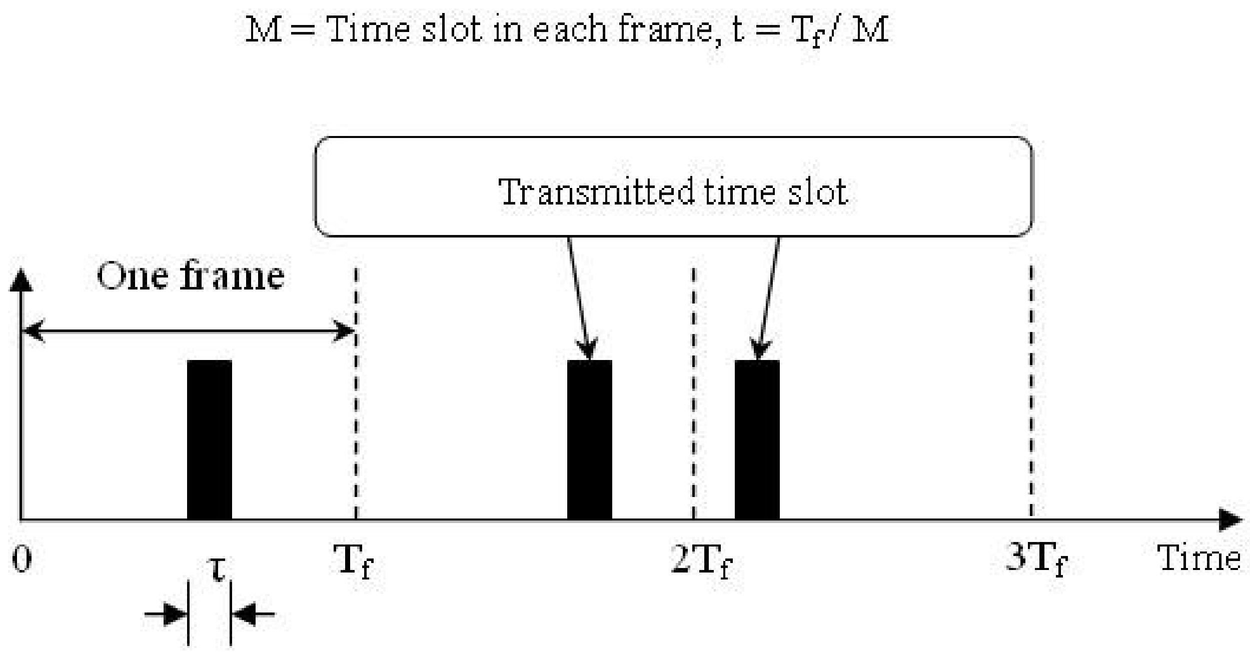


Figure 3.35: Time Hopping Spread Spectrum